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Digital power amplifier, speaker drive system, and speaker drive method

Abstract

An operational amplifier outputs a signal obtained by integrating a sum of a first audio signal that is input to an inverting input terminal and a feedback signal that is output from a power amplifier. A pulse width modulator pulse-width modulates an output of the operational amplifier to generate a PWM signal. A demodulator demodulates the PWM signal to generate a second audio signal that is supplied to a speaker including an audio signal band and a high frequency band of the PWM signal that exceeds the audio signal band. A high pass filter removes low frequency components in a current of the second audio signal including the audio signal band. A current feedback circuit feeds back a current component of the second audio signal from which the low frequency components have been removed, so as to be added to the first audio signal.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7924091	12/2010	Horbach	330/251	H03F 3/217
9647612	12/2016	Watanabe	N/A	H03F 3/2178
9647614	12/2016	Otani	N/A	H03F 3/185
11309853	12/2021	Peterson	N/A	H03F 1/342
2019/0222182	12/2018	Togawa	N/A	H03F 3/217
2020/0153396	12/2019	Nakahira	N/A	H03K 7/08

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2016-072876	12/2015	JP	N/A
2016-072877	12/2015	JP	N/A
2018-088575	12/2017	JP	N/A

OTHER PUBLICATIONS

Written Opinion dated Mar. 22, 2022 issued in corresponding International Application No. PCT/JP2021/046715. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of PCT Application No. PCT/JP2021/046715, filed on Dec. 17, 2021, and claims the priority of Japanese Patent Application No. 2021-041040, filed on Mar. 15, 2021, the entire contents of both of which are incorporated herein by reference.

BACKGROUND

(1) The present disclosure relates to a digital power amplifier, a speaker drive system, and a speaker drive method.

(2) As disclosed in Japanese Unexamined Patent Application Publication No. 2016-72876 (Patent Document 1), a digital power amplifier equipped with a pulse width modulator modulates an input analog audio signal into a digital pulse width modulation signal (PWM signal), amplifies it, and then demodulates it into an analog audio signal to drive a speaker. As an example, in a broadcasting facility where audio signals are supplied to a plurality of speakers to output predetermined sounds, a plurality of parallel-connected speakers may be driven by a plurality of parallel-connected digital power amplifiers.

SUMMARY

(3) When a plurality of digital power amplifiers are connected in parallel, if the voltage gain of each digital power amplifier is not the same, an abnormal current flows between the plurality of digital power amplifiers. In Patent Document 1, the abnormal current flow is prevented by providing a voltage feedback circuit and a current feedback circuit that, respectively, feed back the voltage and current of low frequency components including an audio signal band in the amplified audio signal.

(4) However, the configuration described in Patent Document 1 can only prevent an abnormal current in the audio signal band from flowing. Verification by the present inventor has revealed that an abnormal current flows in a high frequency band that exceeds the audio signal band between a plurality of digital power amplifiers connected in parallel, and that the abnormal current flowing in the high frequency band may cause various problems.

(5) A first aspect of one or more embodiments provides a digital power amplifier including: an operational amplifier in which an analog first audio signal is input to an inverting input terminal; a pulse width modulator configured to generate a pulse width modulation signal based on an output of the operational amplifier; a power amplifier configured to amplify power of the pulse width modulation signal; and a feedback circuit configured to feed back an output of the power amplifier to the inverting input terminal, wherein the operational amplifier is configured to perform a self-oscillating operation by feeding back a feedback signal from the feedback circuit to the inverting input terminal, the pulse width modulator is configured to pulse-width modulate a signal obtained by integrating a sum of the first audio signal and the feedback signal by means of the operational amplifier, thereby generating a pulse width modulation signal for the self-oscillating operation, and the digital power amplifier further includes: a demodulator configured to demodulate the pulse width modulation signal that is output from the power amplifier, and to generate an analog second audio signal that is supplied to a speaker including an audio signal band and a high frequency band of the pulse width modulation signal that exceeds the audio signal band; a high pass filter configured to remove low frequency components including the audio signal band in a current of the second audio signal; and a first current feedback circuit configured to feed back a current component of the second audio signal from which the low frequency components have been removed, so as to be added to the first audio signal.

(6) A second aspect of one or more embodiments provides a speaker drive system including: a plurality of digital power amplifiers connected in parallel, at least one speaker connected to the plurality of digital power amplifiers connected in parallel, wherein each digital power amplifier includes: an operational amplifier in which an analog first audio signal is input to an inverting input terminal; a pulse width modulator configured to generate a pulse width modulation signal based on

an output of the operational amplifier; a power amplifier configured to amplify power of the pulse width modulation signal; and a feedback circuit configured to feed back an output of the power amplifier to the inverting input terminal, wherein the operational amplifier is configured to perform a self-oscillating operation by feeding back a feedback signal from the feedback circuit to the inverting input terminal, the pulse width modulator is configured to pulse-width modulate a signal obtained by integrating a sum of the first audio signal and the feedback signal by means of the operational amplifier, thereby generating a pulse width modulation signal for the self-oscillating operation, and each digital power amplifier further includes: a demodulator configured to demodulate the pulse width modulation signal that is output from the power amplifier, and to generate an analog second audio signal that is supplied to a speaker including an audio signal band and a high frequency band of the pulse width modulation signal that exceeds the audio signal band; a high pass filter configured to remove low frequency components including the audio signal band in a current of the second audio signal; and a first current feedback circuit configured to feed back a current component of the second audio signal from which the low frequency components have been removed, so as to be added to the first audio signal.

(7) A third aspect of one or more embodiments provides a speaker drive method including: inputting an analog first audio signal to an inverting input terminal of an operational amplifier; using a pulse width modulator to generate a pulse width modulation signal based on an output of the operational amplifier; using a power amplifier to amplify power of the pulse width modulation signal; using a feedback circuit to feed back an output of the power amplifier to the inverting input terminal; using the operational amplifier to perform a self-oscillating operation by feeding back a feedback signal from the feedback circuit to the inverting input terminal; using the pulse width modulator to generate a pulse width modulation signal for the self-oscillating operation by performing pulse width modulation on a signal obtained by integrating a sum of the first audio signal and the feedback signal by means of the operational amplifier; using a demodulator to demodulate the pulse width modulation signal that is output from the power amplifier to generate an analog second audio signal that is supplied to a speaker including an audio signal band and a high frequency band of the pulse width modulation signal that exceeds the audio signal band; using a current detection circuit to detect a current of the second audio signal; using a high pass filter to remove low frequency components in a current of the second audio signal including the audio signal band; and using a current feedback circuit to feed back a current component of the second audio signal from which the low frequency components have been removed, so as to be added to the first audio signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a block diagram illustrating a digital power amplifier and a speaker drive system according to one or more embodiments.

(2) FIG. **2** is a characteristic diagram illustrating the relationship between duty ratio and a self-excited frequency when the pulse width modulator **13** shown in FIG. **1** generates a PWM signal.

(3) FIG. **3** is a characteristic diagram illustrating the relationship between the output current Io and the output voltage Vo of the digital power amplifier **100** when the voltage feedback amount β**0** of the voltage feedback circuit **22** and the current feedback amount β**1***a* of the current feedback circuit **24** shown in FIG. **1** are changed.

(4) FIG. **4** is a waveform diagram illustrating a first example of PWM signals generated by the digital power amplifiers **100***a* and **100***b* shown in FIG. **1** and an abnormal current at that time.

(5) FIG. **5** is a waveform diagram illustrating a second example of PWM signals generated by the digital power amplifiers **100***a* and **100***b* shown in FIG. **1** and an abnormal current at that time.

(6) FIG. **6** is characteristic diagram illustrating the relationship between the output current Io and the output voltage Vo of the digital power amplifier **100** supposing that a voltage feedback circuit is provided that feeds back a voltage with a voltage feedback amount β**0***b* in a high frequency band in FIG. **1**, and the voltage feedback amount β**0***b* and the current feedback amount β**1***b* of the voltage feedback circuit **26** are changed.
(7) FIG. **7** is a diagram illustrating frequency bands of an output current component of the digital power amplifier **100** and output current components of the current feedback circuits **24** and **26**.
DETAILED DESCRIPTION
(8) A digital power amplifier, a speaker drive system, and a speaker drive method according to one or more embodiments will be described below with reference to the accompanying drawings.
(9) In FIG. **1**, two digital power amplifiers **100***a* and **100***b* are connected in parallel, and a speaker **50***a* is connected to the digital power amplifiers **100***a* and **100***b*. Any digital power amplifier including the digital power amplifiers **100***a* and **100***b* will be referred to as a digital power amplifier **100**. Three or more digital power amplifiers **100** may be connected in parallel. A plurality of digital power amplifiers **100** connected in parallel constitute a speaker drive system.
(10) Although only one speaker **50***a* is connected to the digital power amplifiers **100***a* and **100***b* in FIG. **1**, a speaker **50***b* may be connected in parallel with the speaker **50***a*. Any speaker including speakers **50***a* and **50***b* will be referred to as a speaker **50**. Three or more speakers **50** may be connected in parallel to the digital power amplifiers **100***a* and **100***b*.
(11) The digital power amplifier **100** shown in FIG. **1** is called a high impedance amplifier. A required number of digital power amplifiers **100** may be connected in parallel according to the number of speakers **50**. At least one speaker **50** may be connected to a plurality of digital power amplifiers **100** connected in parallel.
(12) The configuration and operation of the digital power amplifier **100***b* are the same as the configuration and operation of the digital power amplifier **100***a*. As a representative example, the configuration and operation of the digital power amplifier **100***a* will be described.
(13) As shown in FIG. **1**, the digital power amplifier **100***a* includes an adder **11**, an operational amplifier **12**, a feedback circuit **17**, a pulse width modulator **13**, a power amplifier **14**, a demodulator **15**, a current detection circuit **16**, and a power supply circuit **30**. In addition, the digital power amplifier **100***a* includes a low pass filter **21** (hereinafter, LPF **21**), a voltage feedback circuit **22**, a low pass filter **23** (hereinafter, LPF **23**), a current feedback circuit **24**, a high pass filter **25** (hereinafter, HPF **25**), and a current feedback circuit **26**.
(14) The LPF **21** is a first low pass filter, and the LPF **23** is a second low pass filter. The current feedback circuit **26** is a first current feedback circuit, and the current feedback circuit **24** is a second current feedback circuit.
(15) The power supply circuit **30**, which is connected to an unillustrated commercial power supply, supplies power to the operational amplifier **12**, pulse width modulator **13**, power amplifier **14**, and current detection circuit **16**.
(16) An analog audio signal Sin (a first audio signal) is input to the adder **11**. The adder **11** adds the voltage feedback signal from the voltage feedback circuit **22** and the current feedback signals from the current feedback circuits **24** and **26**, and supplies the result to an inverting input terminal of the operational amplifier **12**. A non-inverting input terminal of the operational amplifier **12** is grounded. The pulse width modulator **13** generates and outputs a digital pulse width modulation signal (PWM signal) having a duty ratio corresponding to the amplitude of the analog signal that is output from the operational amplifier **12**. The pulse width modulator **13** supplies the PWM signal to the power amplifier **14**.
(17) The operational amplifier **12** constitutes an integrating circuit. By feeding back the output of the power amplifier **14** to the inverting input terminal of the operational amplifier **12** by means of the feedback circuit **17**, the operational amplifier **12** performs a self-oscillating operation. The operational amplifier **12** integrates the sum of the audio signal Sin input to the inverting input

terminal and the feedback signal obtained by feeding back the output of the power amplifier **14** to the inverting input terminal by means of the feedback circuit **17**. The pulse width modulator **13** pulse-width modulates the integrated signal to generate a PWM signal for the self-oscillating operation. As long as power is supplied from the power supply circuit **30**, the pulse width modulator **13** generates a PWM signal even if the audio signal Sin is not input to the adder **11**, or the speaker **50** is not connected.
(18) The pulse width modulator **13** generates a PWM signal with a duty ratio of 50% when no analog signal is input from the operational amplifier **12**, that is, when the amplitude of the analog signal is zero. The pulse width modulator **13** generates a PWM signal with a duty ratio increased or decreased from the duty ratio of 50% according to an increase or a decrease in the amplitude of the input analog signal. As shown in FIG. **2** as an example, the self-oscillating pulse width modulator **13** generates a PWM signal with a self-excited frequency according to the duty ratio. In the example shown in FIG. **2**, the self-excited frequency varies from 100 kHz to 500 kHz.
(19) Returning to FIG. **1**, the power amplifier **14** amplifies the power of the PWM signal supplied from the pulse width modulator **13** and supplies it to the demodulator **15**. The demodulator **15** includes a low pass filter (LPF) with an inductor L and a capacitor C. The inductor L is connected in series between the power amplifier **14** and the current detection circuit **16**, and the capacitor C includes one terminal connected to the output stage of the inductor L and the other terminal grounded. The demodulator **15** removes high frequency components from the input digital signal and demodulates it into an analog signal.
(20) The analog signal that is output from the demodulator **15** is supplied to the speaker **50***a* as an analog audio signal Sout (a second audio signal) via the current detection circuit **16**. In this way, the digital power amplifier **100***a* converts the input analog audio signal Sin into a digital PWM signal, amplifies it, and demodulates it into the analog audio signal Sout to drive the speaker **50***a*. As will be described later, the audio signal Sout includes an audio signal band and a high frequency band of the PWM signal exceeding the audio signal band that could not be removed by the demodulator **15**.
(21) The digital power amplifier **100***a* includes the voltage feedback circuit **22** and the current feedback circuit **24** to improve audio performance and circuit stability. Note that improving audio performance includes at least one of suppressing fluctuations in output voltage, reducing a distortion factor, and reducing noise. The LPF **21** removes high frequency components from the voltage of the audio signal Sout after passing through the current detection circuit **16**, and supplies the voltage of the audio signal Sout to the voltage feedback circuit **22**. The voltage feedback circuit **22** feeds back the voltage component from which the high frequency components have been removed, to the adder **11** so as to add it to the audio signal Sin.
(22) The current detection circuit **16** detects the current of the audio signal Sout. The LPF **23** removes the high frequency components of the current of the audio signal Sout detected by the current detection circuit **16**, and supplies the current of the audio signal Sout to the current feedback circuit **24**. The current feedback circuit **24** feeds back the current component from which the high frequency components have been removed, to the adder **11** so as to add it to the audio signal Sin.
(23) Suppose that the amount of voltage feedback by the voltage feedback circuit **22** is β**0**, and the amount of current feedback by the current feedback circuit **24** is β**1***a*, and that the output current of the digital power amplifier **100** (**100***a* and **100***b*) is Io, and the output voltage is Vo. When the voltage feedback amount β**0** and the current feedback amount β**1***a* are changed, the relationship between the output current Io and the output voltage Vo changes as shown in FIG. **3**.
(24) As shown in FIG. **3**, when the current feedback amount β**1***a* is reduced β**0**, for example) and the voltage feedback amount β**0** is increased to cause strong voltage feedback by the voltage feedback circuit **22**, the fluctuation of the output voltage Vo is eliminated and audio performance can be further improved. At this time, the stability of the parallel connection operation of the digital

power amplifiers **100** deteriorates. When the voltage feedback amount β**0** is reduced (0, for example) and the current feedback amount β**1***a* is increased to strongly operate the current feedback by the current feedback circuit **24**, the fluctuation of the output current Io is eliminated and the parallel connection operation of the digital power amplifiers **100** can be improved. At this time, the degree of improvement in audio performance deteriorates.
(25) Thus, the effect of the voltage feedback operation by the voltage feedback circuit **22** and the effect of the current feedback operation by the current feedback circuit **24** are in a trade-off relationship.
(26) When the voltage gains of the digital power amplifiers **100***a* and **100***b* are not the same, and the output voltages Vo of the digital power amplifiers **100***a* and **100***b* are Vo**1** and Vo**2**, respectively, an output current fluctuation width ΔIo occurs in the output current Io. Then, an abnormal current flows between the digital power amplifiers **100***a* and **100***b* in the audio signal band. The larger the slope between the output current Io and the output voltage Vo, the smaller the output current fluctuation width ΔIo, and although the abnormal current in the audio signal band can be suppressed, the degree of improvement in audio performance deteriorates.
(27) Therefore, as shown by the solid line in FIG. **3**, it is preferable to set the slope (ratio of β**1***a*/β**0**) between the output current Io and the output voltage Vo to an appropriate slope (appropriate ratio), by setting the current feedback amount β**1***a* and the voltage feedback amount β**0** to predetermined values exceeding 0. In the digital power amplifier **100**, the optimal value of the ratio of β**1***a*/β**0** is set to achieve both improved audio performance and improved stability of the parallel connection operation of the digital power amplifiers **100** while suppressing the abnormal current in the audio signal band as much as possible.
(28) However, verification by the present inventor has revealed that even if the abnormal current in the audio signal band is suppressed, an abnormal current may flow in the high frequency band exceeding the audio signal band between a plurality of digital power amplifiers **100** connected in parallel, and the abnormal current in the high frequency band may cause various problems. The abnormal current flowing in a high frequency band may cause problems such as deteriorating the efficiency of amplification of the audio signal Sin by the digital power amplifier **100**, deteriorating audio performance, shortening the life of the device by the heating of circuit elements, and increasing the risk of destruction of the device.
(29) In FIG. **4**, (a) and (b) show a first example of PWM signals generated by the digital power amplifiers **100***a* and **100***b*, respectively. The periods of the PWM signals generated by the digital power amplifiers **100***a* and **100***b* are both t**1**, and their frequencies match. However, the PWM signals generated by the digital power amplifiers **100***a* and **100***b* are out of phase.
(30) In this case, an abnormal current shown in (c) of FIG. **4** flows between the digital power amplifiers **100***a* and **100***b* in a high frequency band exceeding the audio signal band. The abnormal current has a period t**1** and alternately flows from the digital power amplifier **100***a* to the digital power amplifier **100***b* and from the digital power amplifier **100***b* to the digital power amplifier **100***a*.
(31) In FIG. **5**, (a) and (b) show a second example of PWM signals generated by the digital power amplifiers **100***a* and **100***b*, respectively. The period of the PWM signal generated by the digital power amplifiers **100***a* is t**1**, the period of the PWM signal generated by the digital power amplifiers **100***b* is t**2**, and their frequencies are different from each other.
(32) In this case as well, an abnormal current shown in (c) of FIG. **5** flows between the digital power amplifiers **100***a* and **100***b* in the high frequency band exceeding the audio signal band. The abnormal current has periods that alternate between a period t**1** and a period t**2**, and alternately flows from the digital power amplifier **100***a* to the digital power amplifier **100***b* and from the digital power amplifier **100***b* to the digital power amplifier **100***a*.
(33) In this way, if a phase or frequency difference exists in the PWM signals generated by the digital power amplifiers **100***a* and **100***b* connected in parallel, an abnormal current flows between

the digital power amplifiers **100** connected in parallel in the high frequency band derived from the PWM signals. As long as power is supplied from the power supply circuit **30**, the abnormal current flows between the digital power amplifiers **100** connected in parallel even if the audio signal Sin is not input to the adder **11**, or even if the speaker **50** is not connected.

(34) Therefore, as shown in FIG. **1**, the digital power amplifier **100***a* is provided with the current feedback circuit **26** for suppressing the abnormal current in the high frequency band derived from the PWM signal. Suppose that the amount of current feedback by the current feedback circuit **26** is β**1***b*. The current of the audio signal Sout detected by the current detection circuit **16** has its low frequency components removed by the HPF **25**, and is fed back to the adder **11** by the current feedback circuit **26**. As described above, the frequency of the PWM signal generated by the pulse width modulator **13** is in the range of 100 kHz to 500 kHz, and the frequency of the abnormal current is the same as that of the PWM signal. The HPF **25** passes a frequency band of 100 kHz to 500 kHz.

(35) In FIG. **1**, there is no voltage feedback circuit that feeds back the voltage in the high frequency band, but suppose that the voltage is fed back with the voltage feedback amount β**0***b* in the high frequency band. When the voltage feedback amount β**0***b* and the current feedback amount β**1***b* are changed, the relationship between the output current Io and the output voltage Vo changes as shown in FIG. **6** similar to FIG. **3**. Similar to the above, if the current feedback by the current feedback circuit **26** is made to be strong by increasing the current feedback amount β**1***b*, the degree of improvement in audio performance is deteriorated, but the stability of the parallel connection operation of the digital power amplifiers **100** can be improved.

(36) However, since there is no audio signal in the high frequency band exceeding the audio signal band, it is not necessary to consider deterioration in the degree of improvement in audio performance. That is, as indicated by the solid line in FIG. **6**, it is only necessary to consider increasing the current feedback amount β**1***b* by the current feedback circuit **26** to improve the stability of the parallel connection operation. Therefore, the current feedback circuit **26** increases the current feedback amount β**1***b* to strongly operate the current feedback. The current feedback amount β**1***b* in the current feedback circuit **26** is larger than the current feedback amount β**1***a* in the current feedback circuit **24**.

(37) Suppose that the PWM signals generated by the digital power amplifiers **100***a* and **100***b* connected in parallel have a phase difference or a frequency difference, and the output voltages Vo of the digital power amplifiers **100***a* and **100***b* are Vo**1** and Vo**2**, respectively. At this time, if the current feedback in the high frequency band is strongly operated, the output current fluctuation width ΔIo can be reduced as shown in FIG. **6**. Therefore, an abnormal current in a high frequency band can be suppressed.

(38) The operation of the digital power amplifier **100** will be described with reference to FIG. **7**. As shown in (a) of FIG. **7**, the output current Io of the digital power amplifier **100** includes an audio signal current component in the audio signal band of 20 Hz to 20 kHz and an abnormal current component of 100 kHz to 500 kHz, for example. An audio signal current component may include an abnormal current in the audio signal band. A current component of the PWM signal is also present between 100 kHz and 500 kHz. Note that the current component and the abnormal current component of the PWM signal are present at one of the frequencies in the frequency range of 100 kHz to 500 kHz.

(39) The LPF **23** removes the high frequency components and supplies a frequency band including the audio signal band of 20 Hz to 20 kHz to the current feedback circuit **24**. Therefore, as shown in (b) of FIG. **7**, the current feedback circuit **24** feeds back the audio signal current component to the adder **11**. The HPF **25** removes the low frequency components and provides a frequency band including 100 kHz to 500 kHz to the current feedback circuit **26**. Therefore, as shown in (c) of FIG. **7**, the current feedback circuit **24** feeds back the abnormal current component of 100 Hz to 500 kHz to the adder **11**.

(40) Due to the current feedback circuit **26** feeds back the abnormal current component to the adder **11**, the phase shift of the PWM signals generated by the digital power amplifiers **100***a* and **100***b* of the self-oscillating PWM modulation system is eliminated, the frequency mismatch is eliminated, and the abnormal current component is suppressed. Therefore, according to the digital power amplifier **100**, it is possible to reduce the occurrence of defects due to the abnormal current flowing in a high frequency band.
(41) The voltage feedback circuit **22** and the current feedback circuit **24** improve audio performance, and suppress the abnormal current in the audio signal band that occurs when the voltage gain of each digital power amplifier **100** is not the same when a plurality of digital power amplifiers **100** are connected in parallel. If the audio performance improvement is not required and the voltage gain can be adjusted and managed so that the voltage gain difference of each digital power amplifier **100** does not occur, the digital power amplifier **100** does not need to be provided with the LPF **21**, voltage feedback circuit **22**, LPF **23**, and current feedback circuit **24**. That is, the digital power amplifier **100** may include only the HPF **25** and current feedback circuit **26** as feedback circuits.
(42) Incidentally, a transformer can be provided at the output of the digital power amplifier **100** to prevent an abnormal current in a high frequency band exceeding the audio signal band from flowing between a plurality of digital power amplifiers **100** connected in parallel. In this case, a large-sized and expensive transformer for high power is required. Therefore, if the digital power amplifier **100** is equipped with a high-power transformer, the device will be large and expensive. According to the configuration of one or more embodiments shown in FIG. **1**, it is possible to prevent abnormal currents flowing in high frequency bands exceeding the audio signal band from flowing between a plurality of digital power amplifiers **100** connected in parallel, without increasing the size of the device and at a low cost.
(43) The present invention is not limited to one or more embodiments described above, and various modifications can be made without departing from the scope of the present invention.

Claims

1. A digital power amplifier comprising: an operational amplifier in which an analog first audio signal is input to an inverting input terminal; a pulse width modulator configured to generate a pulse width modulation signal based on an output of the operational amplifier; a power amplifier configured to amplify power of the pulse width modulation signal; and a feedback circuit configured to feed back an output of the power amplifier to the inverting input terminal, wherein the operational amplifier is configured to perform a self-oscillating operation by feeding back a feedback signal from the feedback circuit to the inverting input terminal, the pulse width modulator is configured to pulse-width modulate a signal obtained by integrating a sum of the first audio signal and the feedback signal by means of the operational amplifier, thereby generating a pulse width modulation signal for the self-oscillating operation, and the digital power amplifier further comprises: a demodulator configured to demodulate the pulse width modulation signal that is output from the power amplifier, and to generate an analog second audio signal that is supplied to a speaker including an audio signal band and a high frequency band of the pulse width modulation signal that exceeds the audio signal band; a high pass filter configured to remove low frequency components including the audio signal band in a current of the second audio signal; and a first current feedback circuit configured to feed back a current component of the second audio signal from which the low frequency components have been removed, so as to be added to the first audio signal.
2. The digital power amplifier according to claim 1, further comprising: a first low pass filter configured to remove high frequency components including the high frequency band in a voltage of the second audio signal; a voltage feedback circuit configured to feed back a voltage component of

the second audio signal from which the high frequency components have been removed, so as to be added to the first audio signal; a second low pass filter configured to remove high frequency components including the high frequency band in a current of the second audio signal; a second current feedback circuit configured to feed back a current component of the second audio signal from which the high frequency components have been removed, so as to be added to the first audio signal, wherein an amount of current feedback performed by the first current feedback circuit is larger than an amount of current feedback performed by the second current feedback circuit.

3. A speaker drive system comprising: a plurality of digital power amplifiers connected in parallel, at least one speaker connected to the plurality of digital power amplifiers connected in parallel, wherein each digital power amplifier comprises: an operational amplifier in which an analog first audio signal is input to an inverting input terminal; a pulse width modulator configured to generate a pulse width modulation signal based on an output of the operational amplifier; a power amplifier configured to amplify power of the pulse width modulation signal; and a feedback circuit configured to feed back an output of the power amplifier to the inverting input terminal, wherein the operational amplifier is configured to perform a self-oscillating operation by feeding back a feedback signal from the feedback circuit to the inverting input terminal, the pulse width modulator is configured to pulse-width modulate a signal obtained by integrating a sum of the first audio signal and the feedback signal by means of the operational amplifier, thereby generating a pulse width modulation signal for the self-oscillating operation, and each digital power amplifier further comprises: a demodulator configured to demodulate the pulse width modulation signal that is output from the power amplifier, and to generate an analog second audio signal that is supplied to a speaker including an audio signal band and a high frequency band of the pulse width modulation signal that exceeds the audio signal band; a high pass filter configured to remove low frequency components including the audio signal band in a current of the second audio signal; and a first current feedback circuit configured to feed back a current component of the second audio signal from which the low frequency components have been removed, so as to be added to the first audio signal.

4. A speaker drive method comprising: inputting an analog first audio signal to an inverting input terminal of an operational amplifier; using a pulse width modulator to generate a pulse width modulation signal based on an output of the operational amplifier; using a power amplifier to amplify power of the pulse width modulation signal; using a feedback circuit to feed back an output of the power amplifier to the inverting input terminal; using the operational amplifier to perform a self-oscillating operation by feeding back a feedback signal from the feedback circuit to the inverting input terminal; using the pulse width modulator to generate a pulse width modulation signal for the self-oscillating operation by performing pulse width modulation on a signal obtained by integrating a sum of the first audio signal and the feedback signal by means of the operational amplifier; using a demodulator to demodulate the pulse width modulation signal that is output from the power amplifier to generate an analog second audio signal that is supplied to a speaker including an audio signal band and a high frequency band of the pulse width modulation signal that exceeds the audio signal band; using a current detection circuit to detect a current of the second audio signal; using a high pass filter to remove low frequency components in a current of the second audio signal including the audio signal band; and using a current feedback circuit to feed back a current component of the second audio signal from which the low frequency components have been removed, so as to be added to the first audio signal.